

SMAP CDF-matching sensitivity experiment over North Morocco

Domain: NorthMor | Year: 2016

Objective:

This report evaluates the internal behaviour of the SMAP EnKF assimilation system over NorthMor in 2016. It focuses on SMAP observation availability, innovations, analysis increments, seasonal correction patterns, and ensemble spread diagnostics for the noCDF and CDF assimilation configurations.

Experiments Compared:

- OPL_noirr_2016
- DA_smap_nocdf_noirr_2016
- DA_smap_cdf_noirr_2016

Note: This report does not include hydrological impact diagnostics. Soil moisture, ET, runoff, groundwater, streamflow, and external validation diagnostics were not generated in this report version.

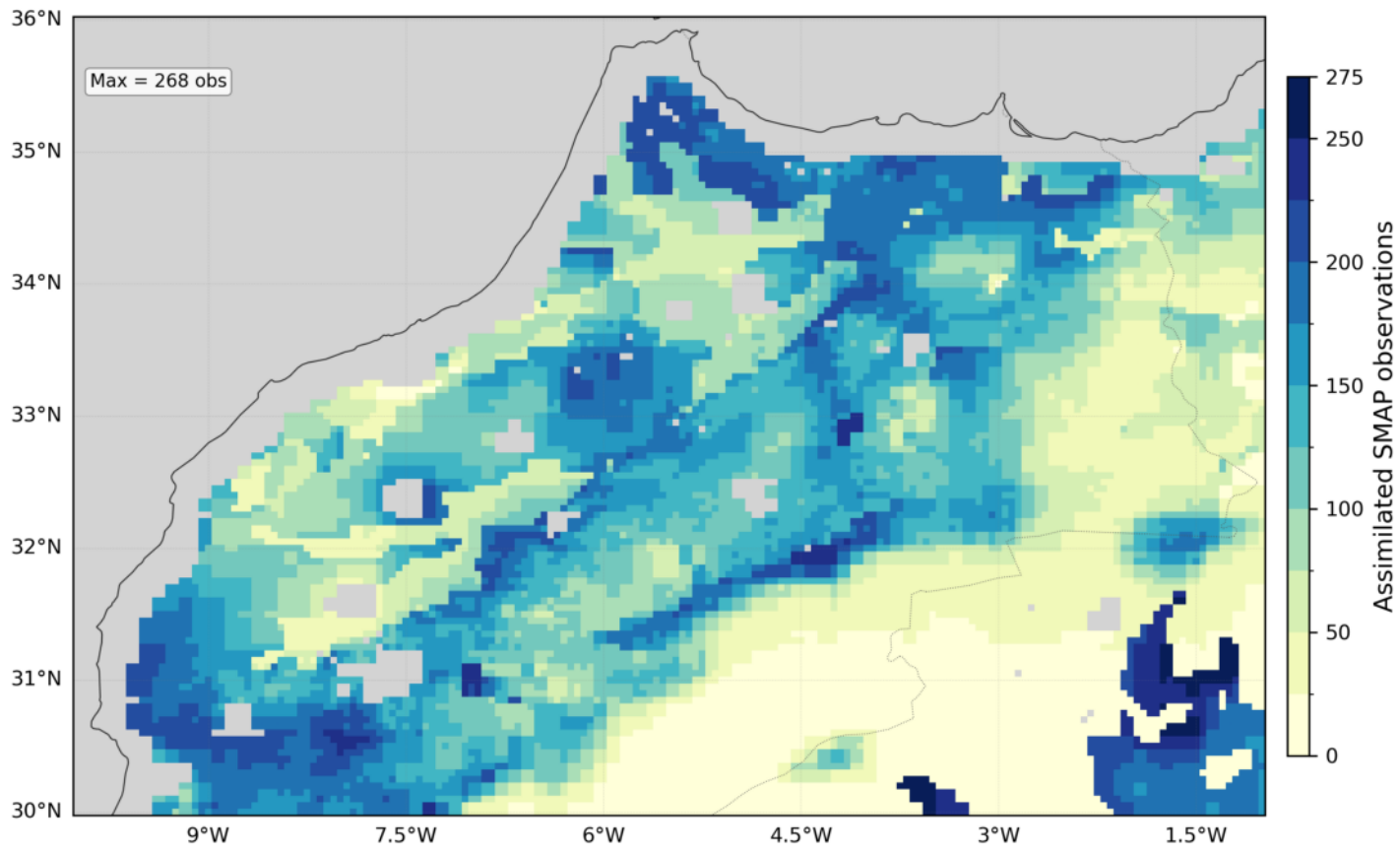
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2. DA-SMAP-noCDF assimilation diagnostics

Spatial distribution of assimilated SMAP observations — DA-SMAP-noCDF, NorthMor, 2016

Spatial distribution of assimilated SMAP observations — DA-SMAP-noCDF, NorthMor, 2016



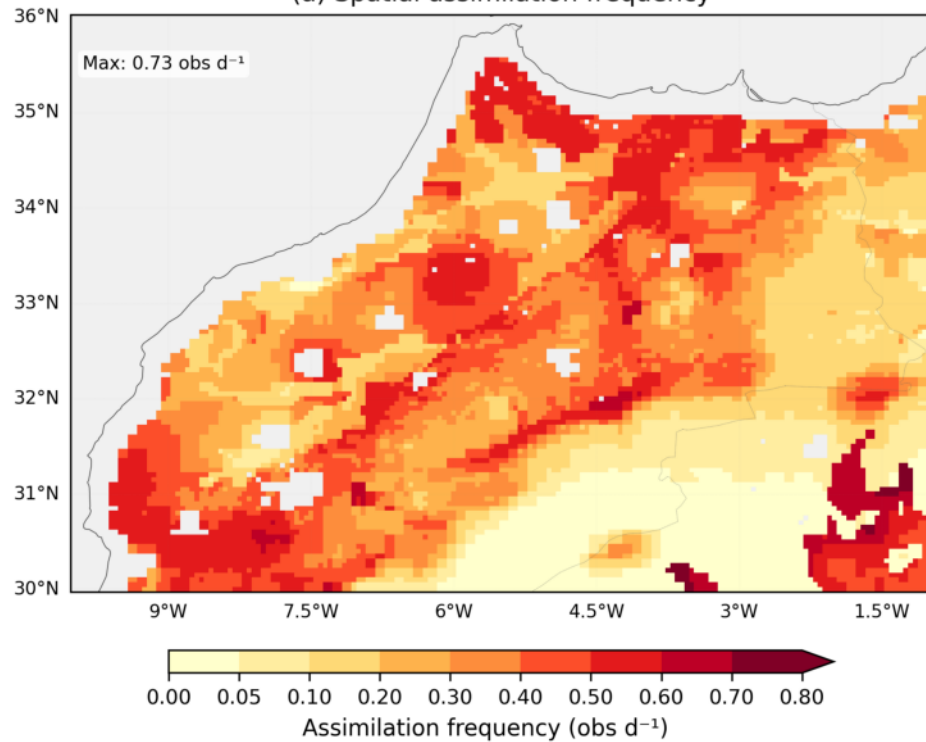
Source file: DA_smap_nocdf_noirr_2016_obs_count.png

Number of assimilated SMAP observations per grid cell during January-December 2016. Higher values indicate more frequent valid SMAP updates.

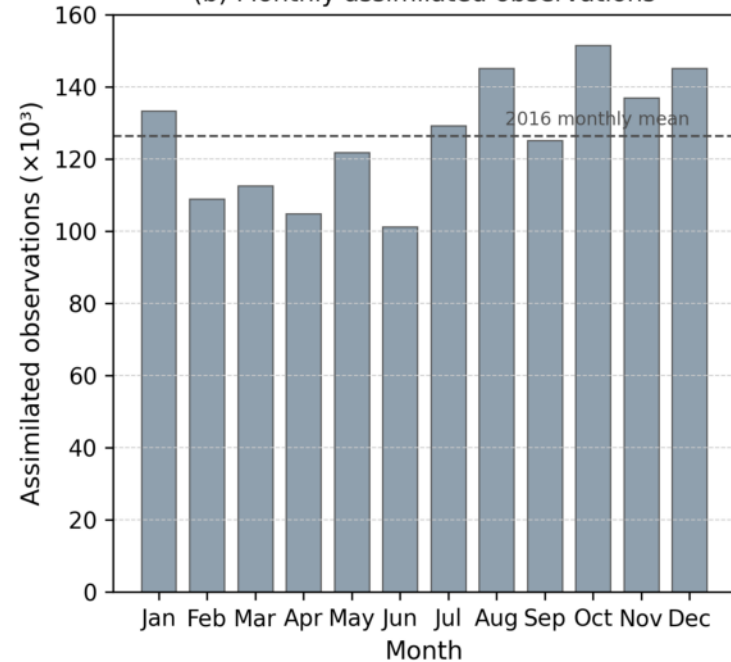
Monthly SMAP assimilation frequency — DA-SMAP-noCDF, NorthMor, 2016

Monthly SMAP assimilation frequency — DA-SMAP-noCDF, NorthMor, 2016

(a) Spatial assimilation frequency



(b) Monthly assimilated observations

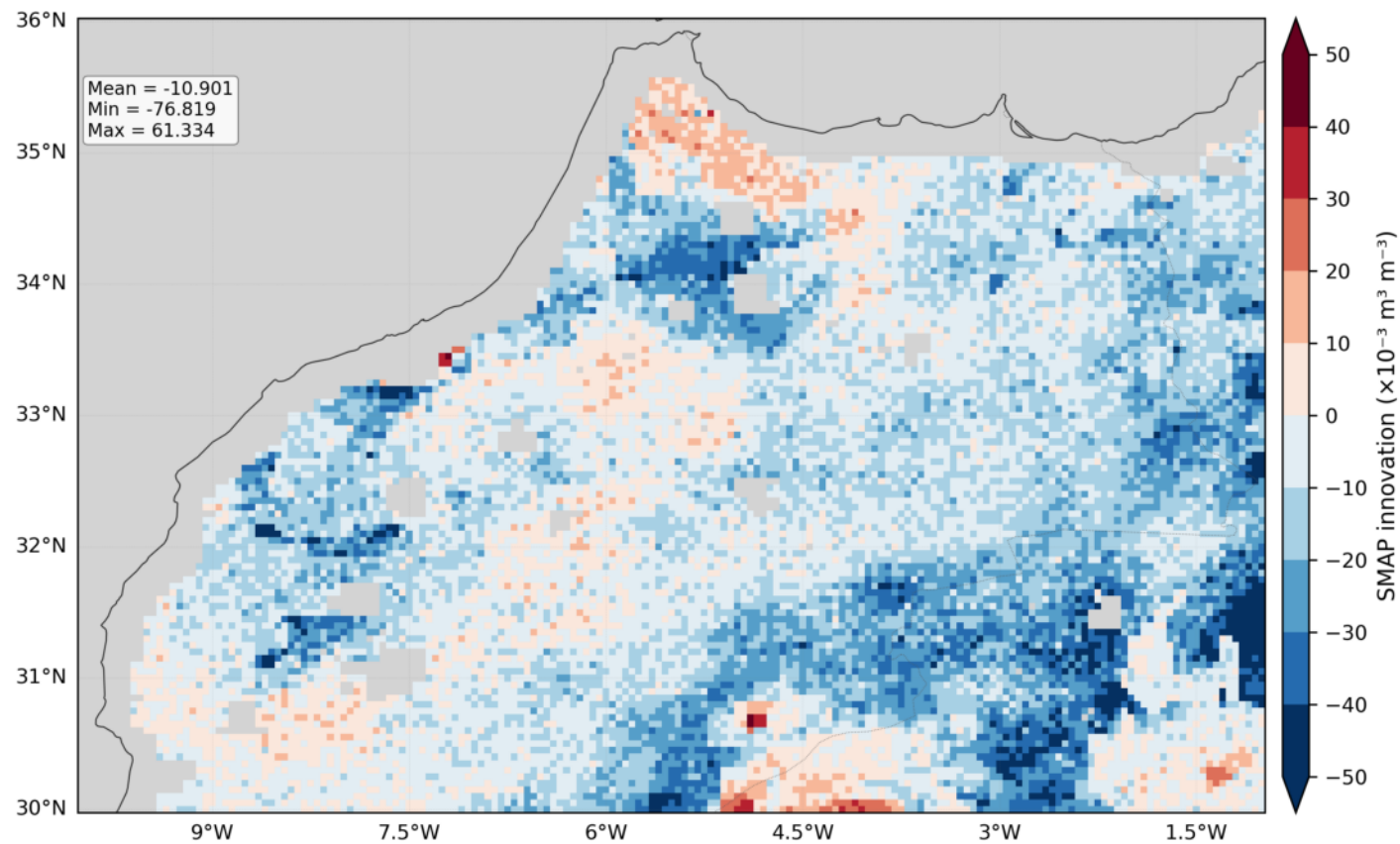


Source file: DA_smap_nocdf_noirr_2016_assim_frequency.png

Spatial assimilation frequency and monthly total number of assimilated observations. This diagnostic identifies spatial and temporal gaps in SMAP availability.

Mean SMAP innovation, observation minus forecast — DA-SMAP-noCDF, NorthMor, 2016

Mean SMAP innovation, observation minus forecast — DA-SMAP-noCDF, NorthMor, 2016

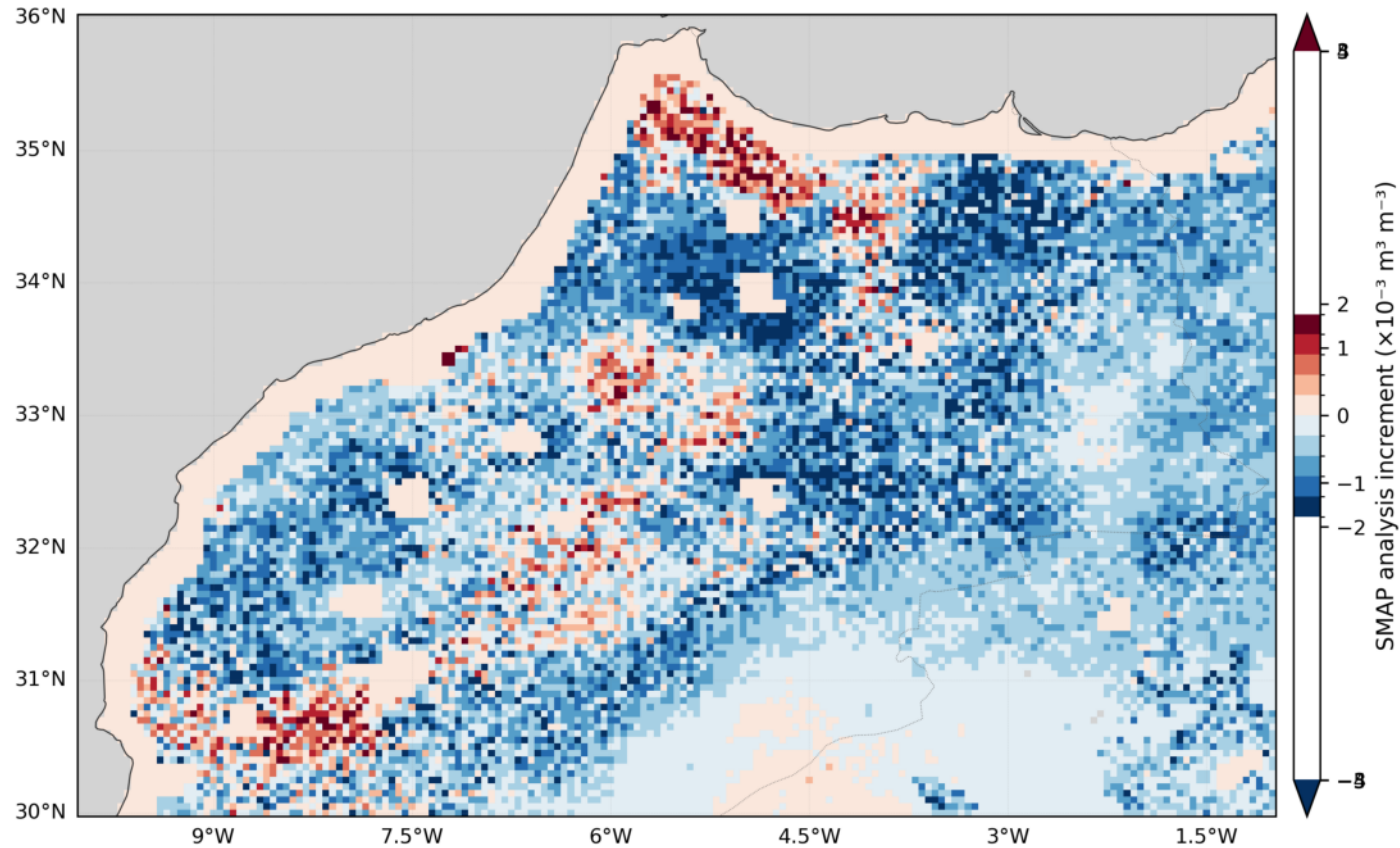


Source file: DA_smap_nocdf_noirr_2016_innovation_map.png

Mean innovation computed as observation minus forecast. Positive values indicate that SMAP is wetter than the model forecast; negative values indicate that SMAP is drier.

Mean SMAP analysis increment, analysis minus forecast — DA-SMAP-noCDF, NorthMor, 2016

Mean SMAP analysis increment, analysis minus forecast — DA-SMAP-noCDF, NorthMor, 2016

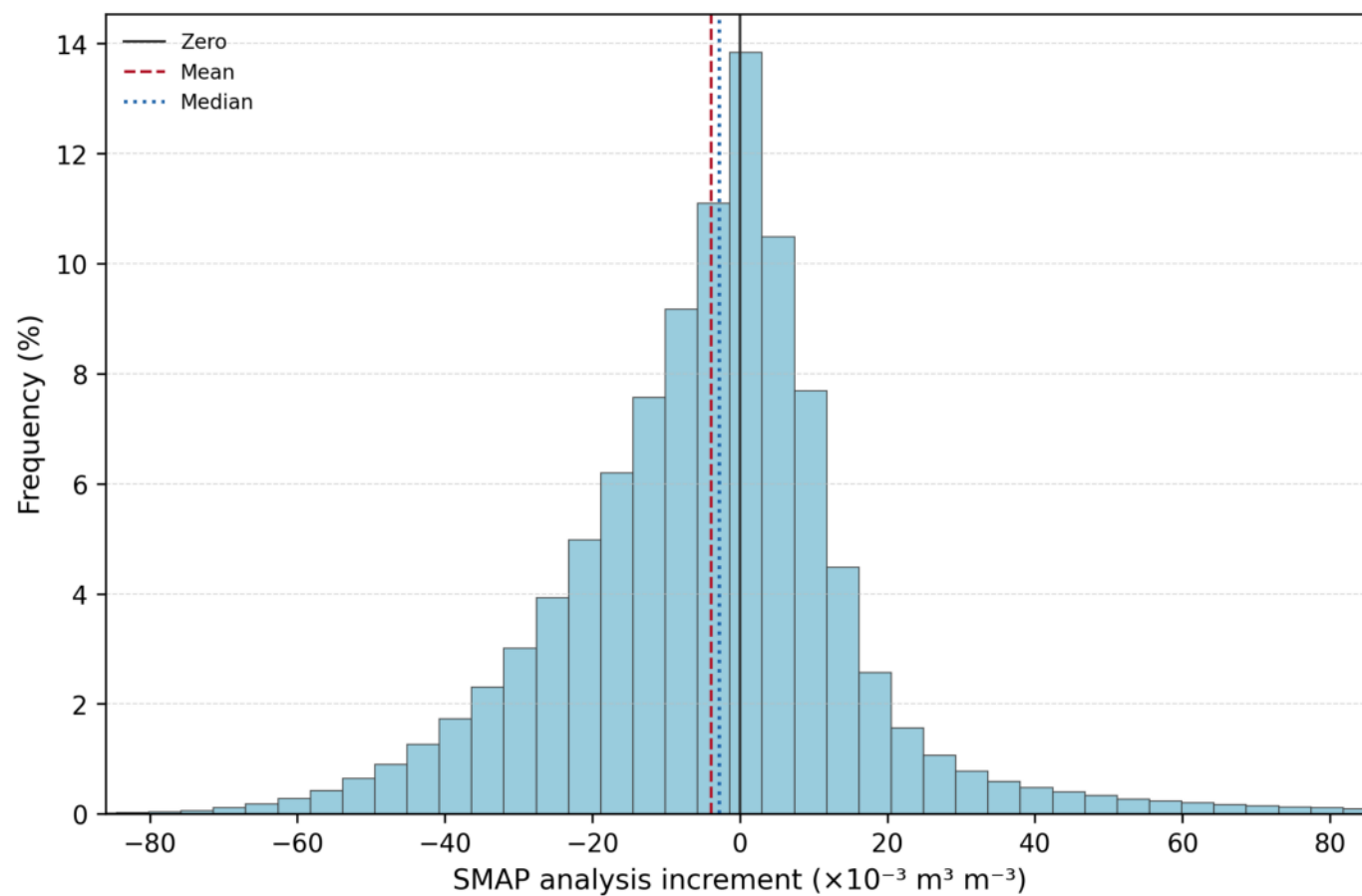


Source file: DA_smap_nocdf_noirr_2016_increment_map.png

Mean analysis increment computed as analysis minus forecast. Positive values indicate wetting corrections; negative values indicate drying corrections applied by the EnKF.
Colorbar limits: robust symmetric bounds [-1.5e-3, 1.5e-3].

Distribution of SMAP analysis increments — DA-SMAP-noCDF, NorthMor, 2016

Distribution of SMAP analysis increments — DA-SMAP-noCDF, NorthMor, 2016

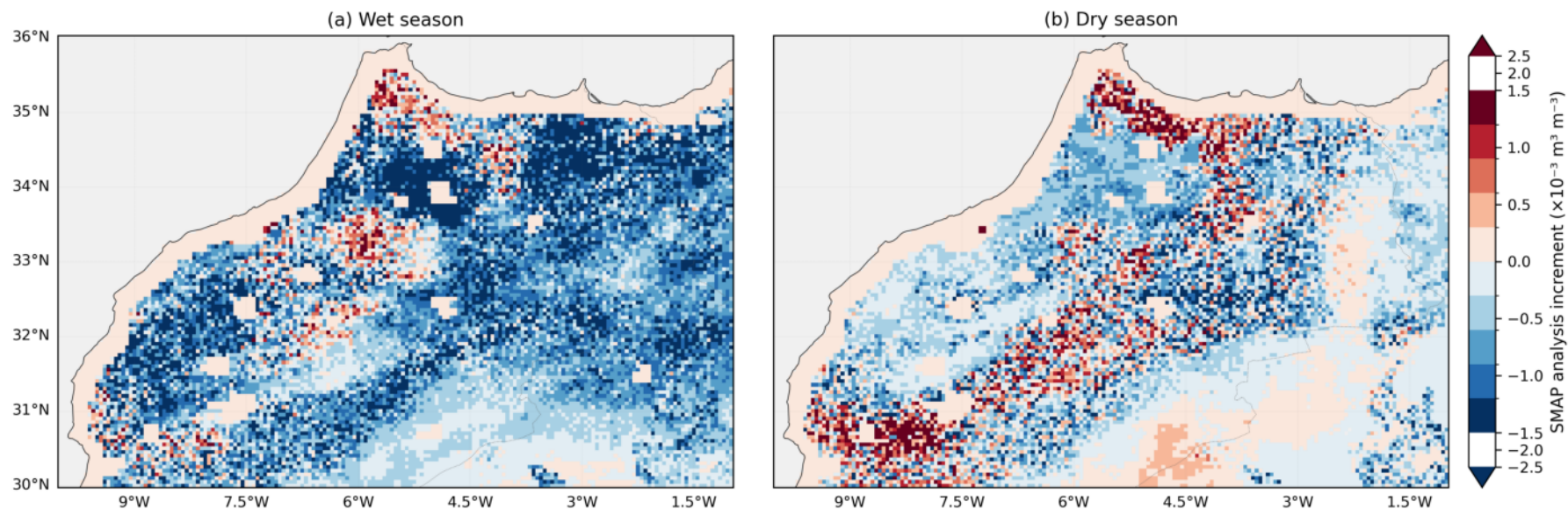


Source file: DA_smap_nocdf_noirr_2016_increment_histogram.png

Distribution of all SMAP analysis increments over the domain and year. The mean and median indicate whether the assimilation applies a systematic wetting or drying correction.

Wet- and dry-season SMAP analysis increments — DA-SMAP-noCDF, NorthMor, 2016

Wet- and dry-season SMAP analysis increments — DA-SMAP-noCDF, NorthMor, 2016



Source file: DA_smap_nocdf_noirr_2016_seasonal_increments.png

Seasonal mean increments during wet and dry seasons. This diagnostic shows whether the assimilation correction depends on seasonal hydroclimatic conditions.

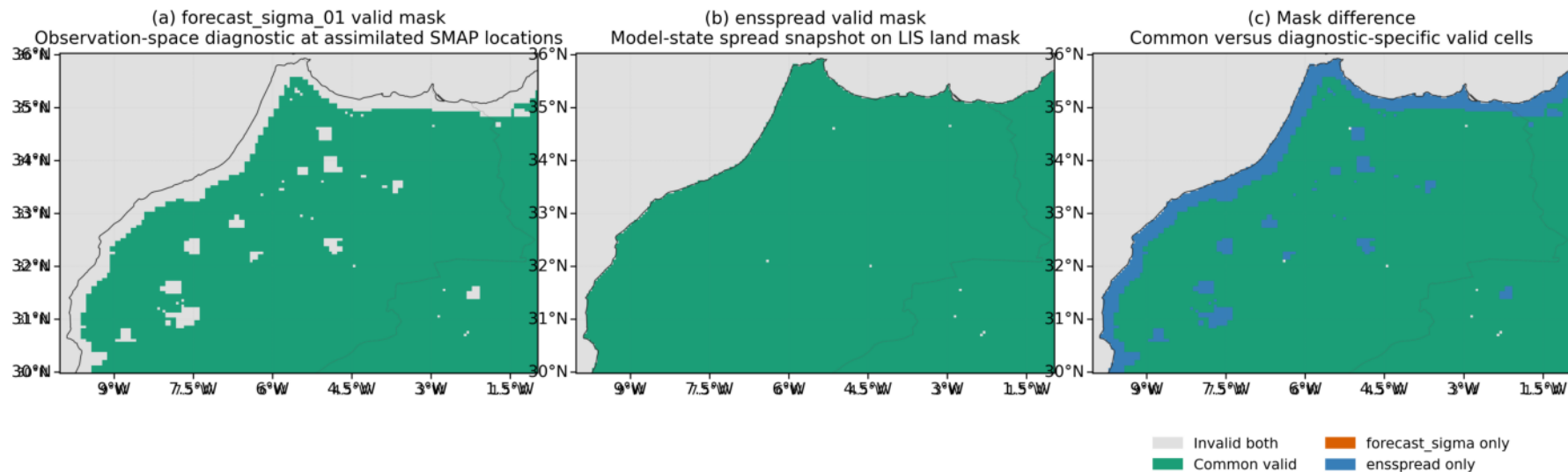
Wet season = Nov, Dec, Jan, Feb, Mar, Apr

Dry season = May, Jun, Jul, Aug, Sep, Oct

Colorbar limits: robust symmetric bounds $[-1.5\text{e-}3, 1.5\text{e-}3]$.

Spread diagnostic consistency check — DA-SMAP-noCDF, NorthMor, 2016

Spread diagnostic consistency check — DA-SMAP-noCDF, NorthMor, 2016



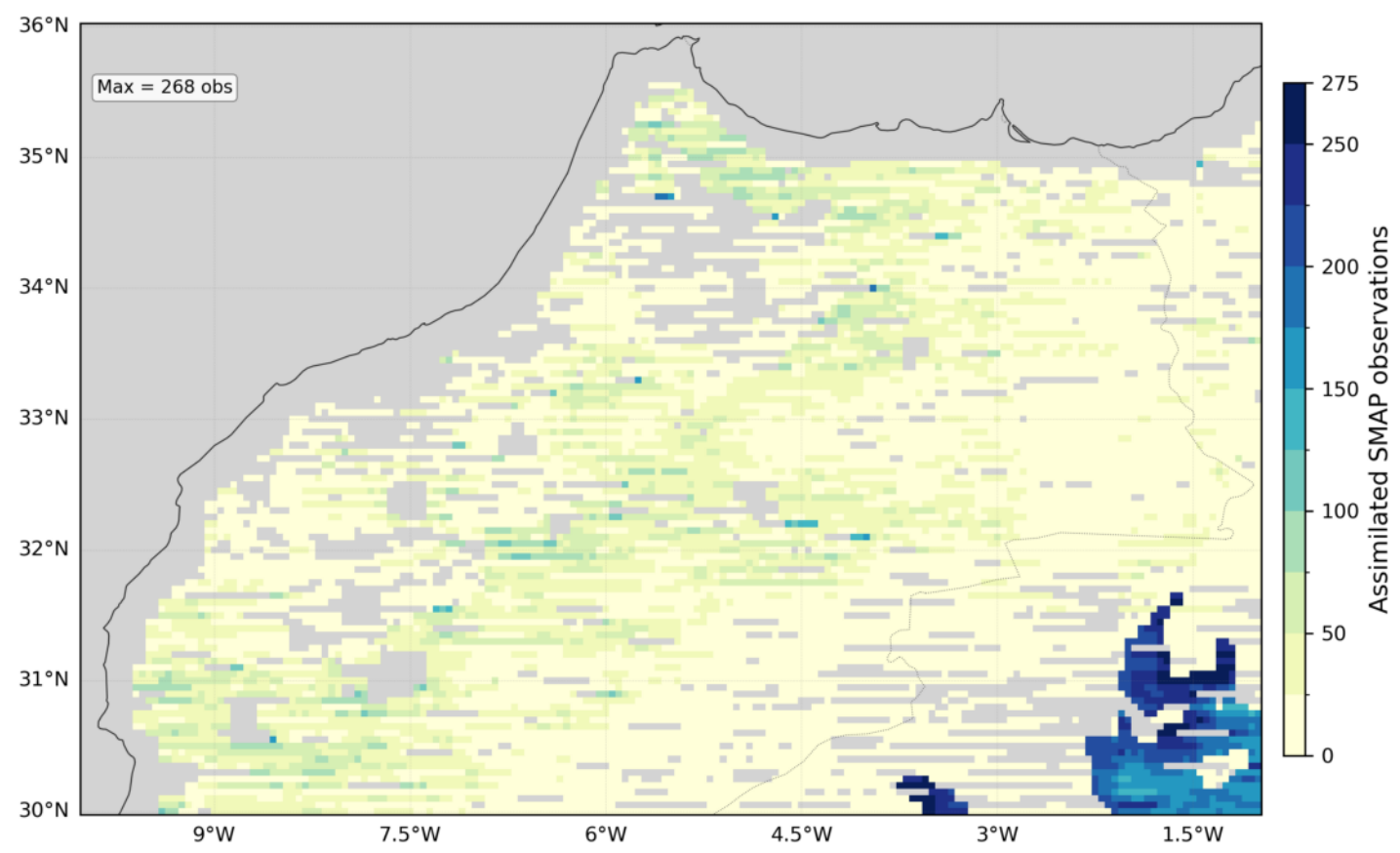
Source file: DA_smap_nocdf_noirr_2016_spread_consistency.png

Technical quality-control diagnostic used to verify the consistency of masks, observation-space uncertainty fields, and model-state spread fields. This figure should not be interpreted as hydrological validation.

3. DA-SMAP-CDF assimilation diagnostics

Spatial distribution of assimilated SMAP observations — DA-SMAP-CDF, NorthMor, 2016

Spatial distribution of assimilated SMAP observations — DA-SMAP-CDF, NorthMor, 2016



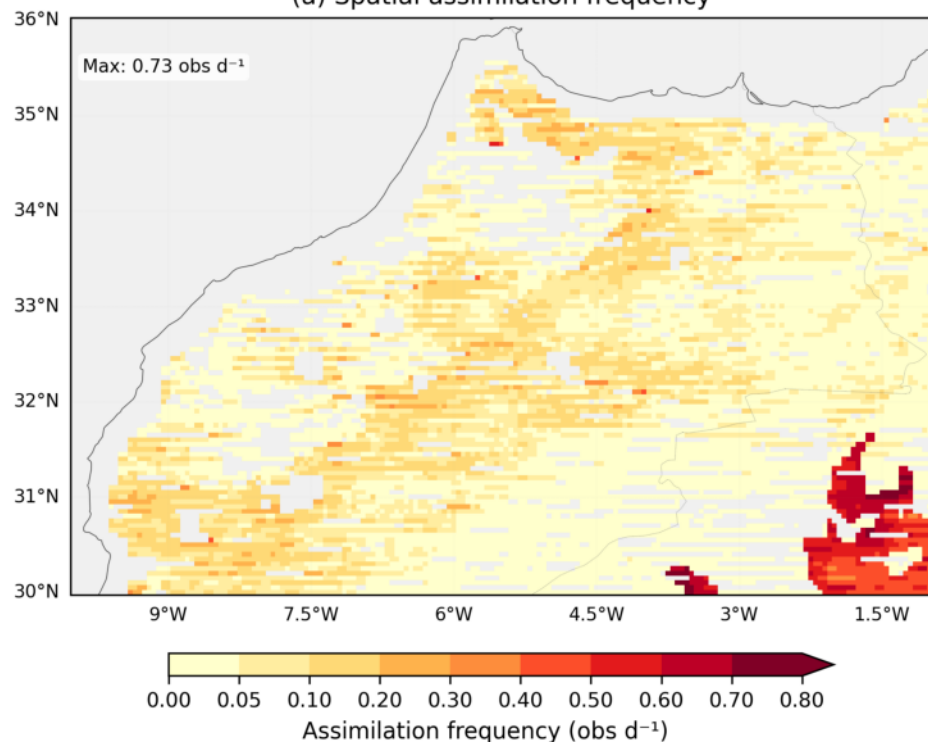
Source file: DA_smap_cdf_noirr_2016_obs_count.png

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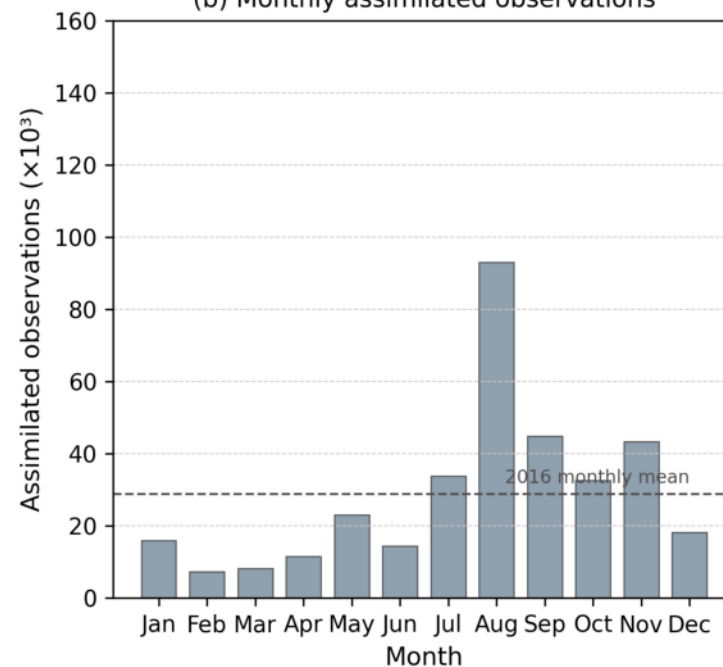
Monthly SMAP assimilation frequency — DA-SMAP-CDF, NorthMor, 2016

Monthly SMAP assimilation frequency — DA-SMAP-CDF, NorthMor, 2016

(a) Spatial assimilation frequency



(b) Monthly assimilated observations

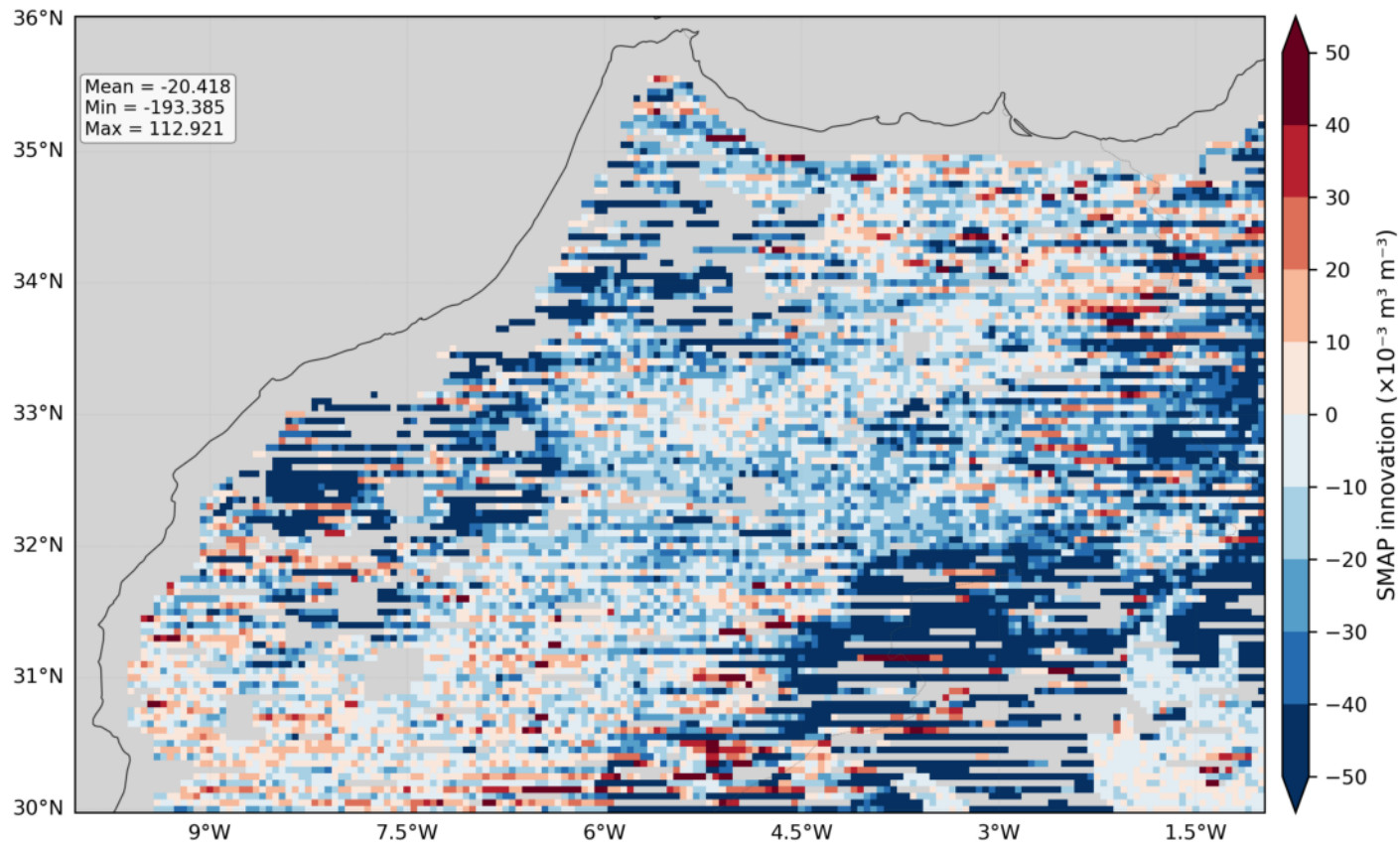


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Spatial assimilation frequency and monthly total number of assimilated observations. This diagnostic identifies spatial and temporal gaps in SMAP availability.

Mean SMAP innovation, observation minus forecast — DA-SMAP-CDF, NorthMor, 2016

Mean SMAP innovation, observation minus forecast — DA-SMAP-CDF, NorthMor, 2016

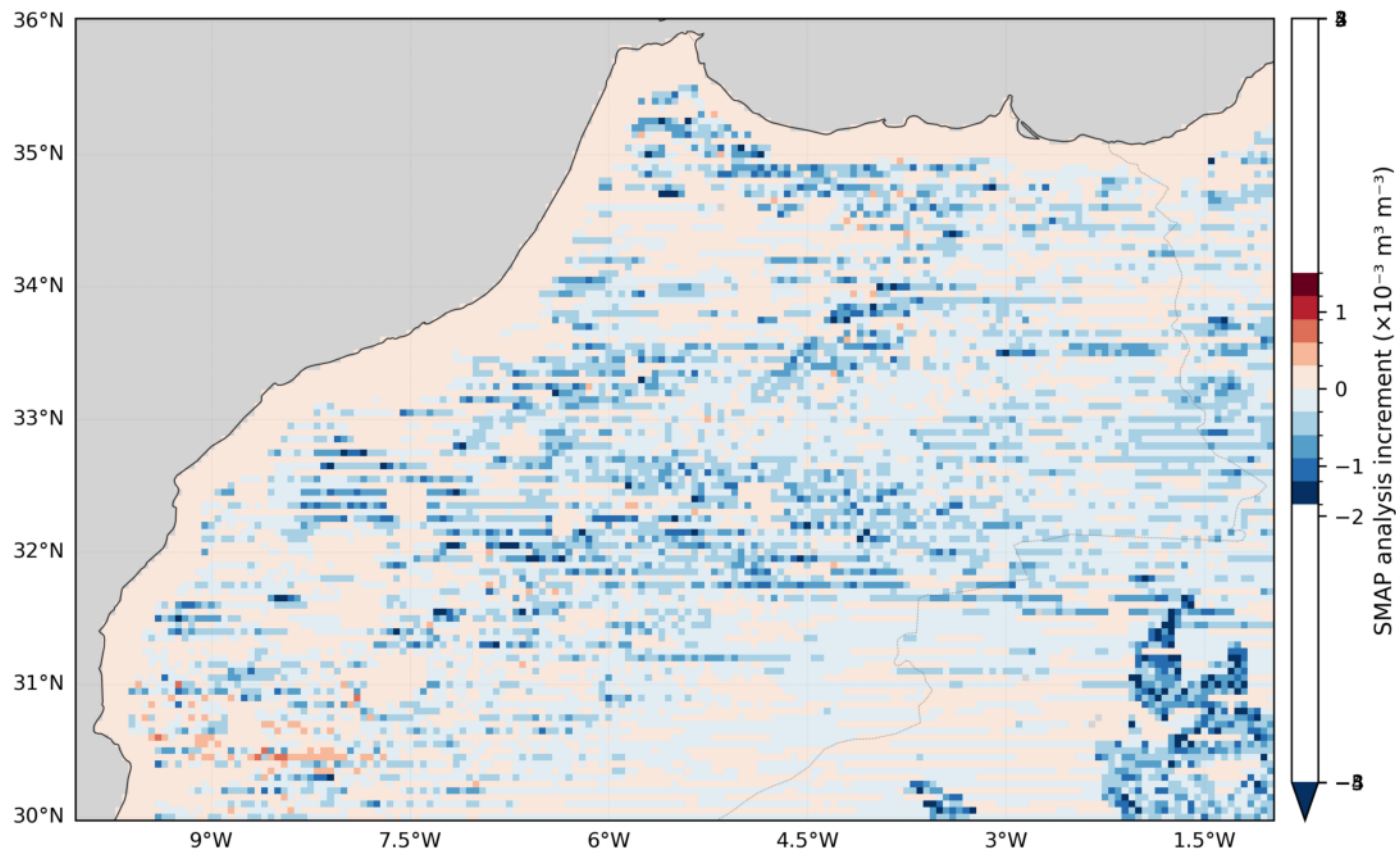


Source file: DA_smap_cdf_noirr_2016_innovation_map.png

Mean innovation computed as observation minus forecast. Positive values indicate that SMAP is wetter than the model forecast; negative values indicate that SMAP is drier.

Mean SMAP analysis increment, analysis minus forecast — DA-SMAP-CDF, NorthMor, 2016

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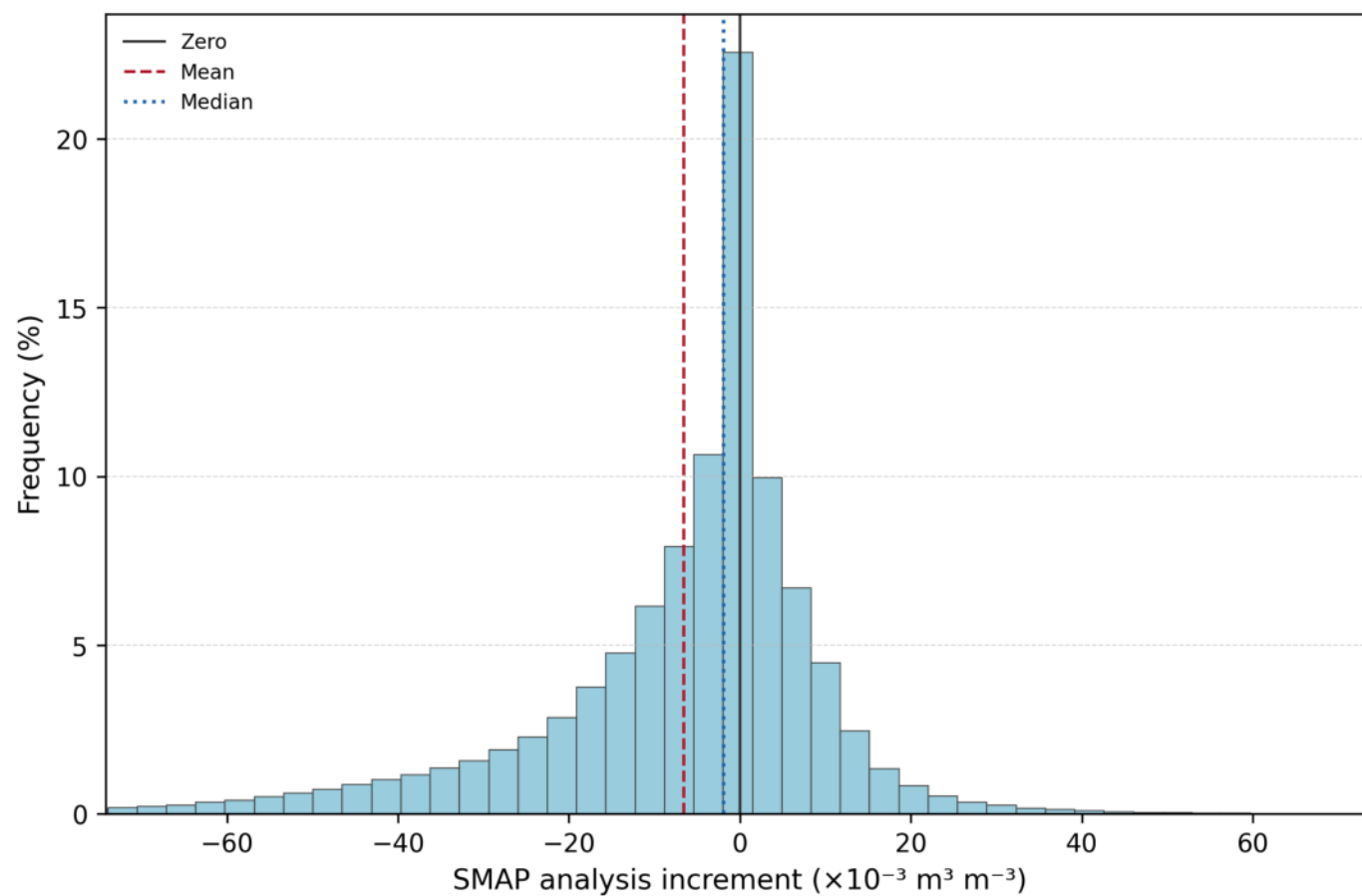


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Mean analysis increment computed as analysis minus forecast. Positive values indicate wetting corrections; negative values indicate drying corrections applied by the EnKF.
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Distribution of SMAP analysis increments — DA-SMAP-CDF, NorthMor, 2016

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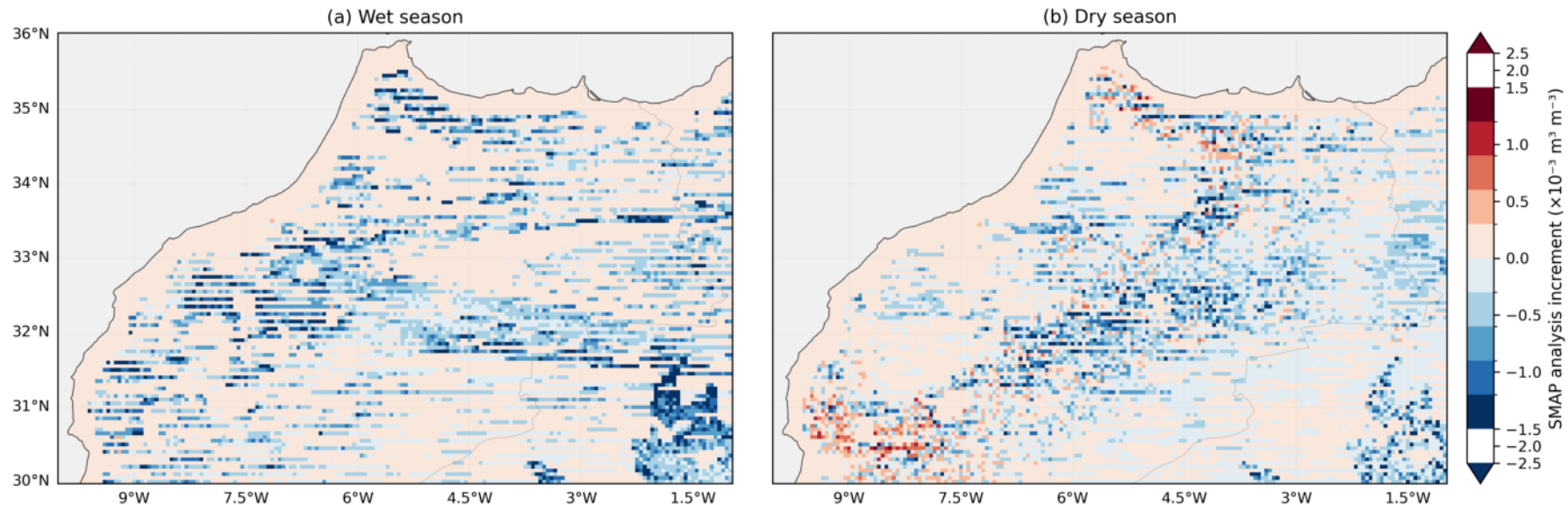


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Distribution of all SMAP analysis increments over the domain and year. The mean and median indicate whether the assimilation applies a systematic wetting or drying correction.

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Seasonal mean increments during wet and dry seasons. This diagnostic shows whether the assimilation correction depends on seasonal hydroclimatic conditions.

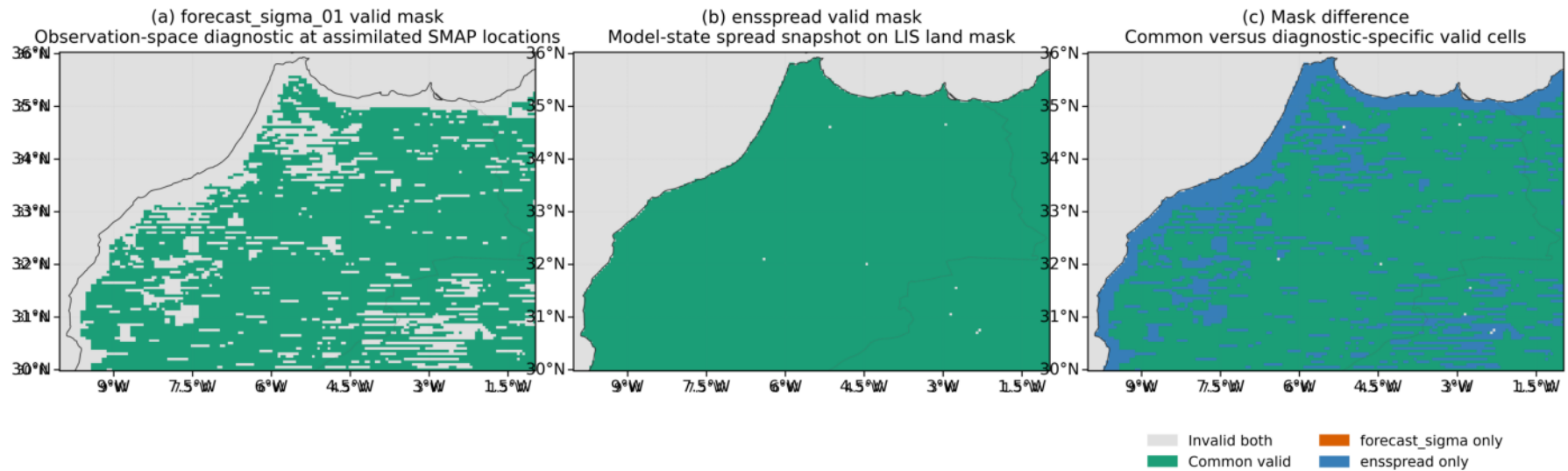
Wet season = Nov, Dec, Jan, Feb, Mar, Apr

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Spread diagnostic consistency check — DA-SMAP-CDF, NorthMor, 2016

Spread diagnostic consistency check — DA-SMAP-CDF, NorthMor, 2016



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Technical quality-control diagnostic used to verify the consistency of masks, observation-space uncertainty fields, and model-state spread fields. This figure should not be interpreted as hydrological validation.

4. noCDF vs CDF interpretation notes

When comparing noCDF and CDF assimilation diagnostics, check:

- whether the spatial coverage of assimilated observations is consistent between experiments;
- whether CDF-matching reduces systematic innovation bias;
- whether the increment distribution becomes more centred around zero;
- whether wet-season and dry-season increments show coherent spatial patterns;
- whether spread diagnostics remain consistent;
- whether any difference in observation count reflects a real processing choice or a labelling/input-file issue.

Do not invent scientific conclusions. This page should only guide the interpretation.

5. Diagnostics not included in this report

This PDF focuses only on the SMAP assimilation diagnostics already generated for the 2016 CDF-matching sensitivity experiment.

Hydrological impact diagnostics such as soil moisture response, ET, runoff, groundwater, streamflow, and external validation were not requested in this report version and are therefore not included.

6. Scientific checklist

- ☐ Are SMAP observations spatially well distributed?
- ☐ Are there spatial gaps in the assimilation coverage?
- ☐ Is the monthly observation availability realistic?
- ☐ Are the noCDF and CDF observation counts consistent?
- ☐ Are innovations mostly positive, negative, or spatially mixed?
- ☐ Are increments physically coherent with the sign of innovations?
- ☐ Is the increment distribution centred close to zero?
- ☐ Does CDF-matching reduce systematic innovation bias?
- ☐ Are wet-season and dry-season increments different?
- ☐ Are there regions with unrealistic or noisy increments?
- ☐ Are spread diagnostics consistent enough for interpretation?
- ☐ Which assimilation figures should be kept for the paper?
- ☐ Which assimilation figures should go to supplementary material?
- ☐ Which figures need revised colorbar limits, titles, or captions?